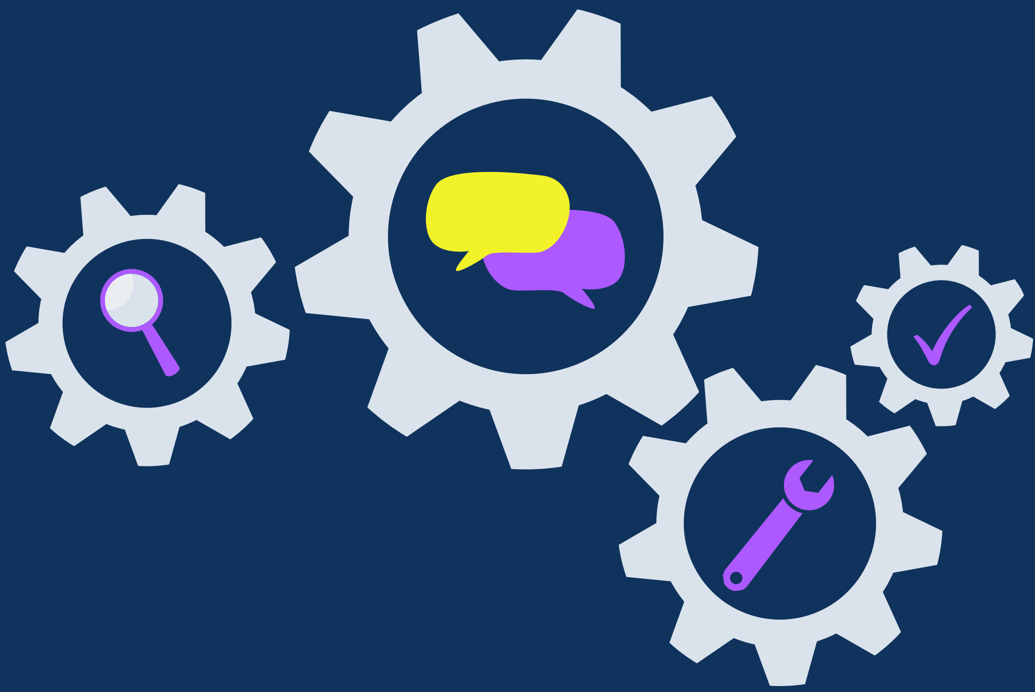
AulaSputnik

An educational project aimed at inspiring values through innovative workshops



aula SPUTNIK

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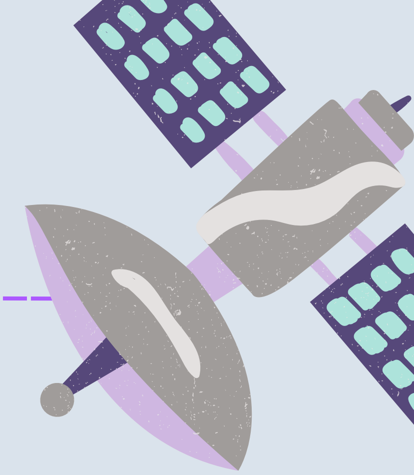

A project to teach values using space as a context

Context

In the context of the new society we are building—an uncertain and complex one—it is crucial to help people develop and unlock new abilities to learn how to cooperate and participate. For this reason, a community needs to embrace and build itself through the scientific method, critical thinking, and inspiration.

The world of science and technology is a distant world for many people. This is why humanizing science and technology is a fundamental challenge, especially in the aerospace sector. The aerospace sector is a field that drives the improvement of academic and employability skills, while also innovating to find solutions to new social challenges such as climate change.

For this reason, education and the aerospace sector are two of the keys to making progress on these future collective challenges. The new generations are the key to building a more cooperative, leading, and inclusive society, where anyone—regardless of gender, background, age, or culture—can have the same opportunities.

Objetive

AulaSputnik is an educational project focused on organizing activities designed to inspire and spark the talents of future generations, using the aerospace sector as a framework, in a way that is inclusive, accessible, and participatory for everyone.

We believe in developing scientific and technological competencies—understood as a fundamental (and therefore essential) skill—for all people, grounded in core values.

Mission

The mission is to promote positive attitudes, emotions, and identities related to science and society; the target audience is children aged 8 to 14. We will explore the importance of social participation and leadership in science and enhance academic and employability skills, using the aerospace sector as a context.

Vision

Our vision is to spark curiosity and creativity, as well as to inspire personal skills and teamwork. We aim to inspire, educate and expanding opportunities to engage with science.

Educational workshops

An innovative educational program designed to open up new skills and perspectives

Methodology

The workshop sessions follow a structure that begins with an INTRODUCTION to the CHALLENGE and leads to a CONCLUSION.

Sessions last 1.5 HOURS for groups of 15-20 people. A resource packet is provided for both participants and instructors.

CHALLENGE → CONCLUSION

List of workshops

- **Workshop 1: How do nanosatellites help our planet?**
- **Workshop 2: Explore the space with AulaSPUTNIK rockets!**
- **Workshop 3: The women in the aerospace sector**
- **Workshop 4: Constellations as a life notebook**
- **Workshop 5: Biology in space: What do aliens look like?**
- **Workshop 6: Create your own star system!**

Workshop 1: How do nanosatellites help our planet?

Key Features

- Age range: 8–14 years
- Type: group activities
- Difficulty: moderate
- Duration: 1 hour 30 minutes
- Materials included: building materials, pencil, rubbers, and activity booklet

Summary

The main objective of this workshop is to explain what satellites (and specifically nanosatellites) are and to explain the basic principles of this technology, in order to understand how they can help us improve our planet.

Next, through a group challenge, students must design a satellite, build a model using a basic kit, and give a presentation on their project idea. The purpose of this satellite must be to provide a service that helps our society reduce the impact of climate change. Therefore, students will need to use their creativity and ability to find solutions to various social challenges.

The students will learn...

- The phases of a space mission: satellite design, construction, and launch.
- An assessment of the impact of technological development on living and working conditions.
- The advancements, products, and materials that contribute to societal progress.
- Awareness of relevant social issues

Skills to be developed...

- Oral communication.
- Teamwork.
- Creativity in proposing imaginative and original solutions.
- STEAM skills.

Workshop 2: Explore the space with AulaSPUTNIK rockets!

Key Features

- Age range: 8–14 years
- Type: group activities
- Difficulty: moderate
- Duration: 1 hour 30 minutes
- Materials included: workbook, pencils, rubbers, scissors, glue, tape, empty plastic bottles, and 3D-printed parts

Summary

The main objective of this workshop is to explain what rockets are, how they work, and why they are so important for exploring space and learning more about the world around us.

Next, through a group challenge, students will be presented with a set of hypotheses with the goal of designing and building a paper rocket and a launch pad. They will need to coordinate and organize themselves into roles and tasks. Afterward, the rockets will be launched to answer these questions and collect the necessary distance data. Based on this exercise, each team will reflect on the conclusions drawn from their results and give a presentation on their project.

The students will learn...

- Analysis of the challenges involved in leaving Earth's atmosphere.
- The ability to evaluate and make logical decisions in an impartial and reasoned manner.
- The phases of a space mission: rocket design, construction, and launch.
- Planning and organization through a simulated space mission.

Skills to be developed...

- Oral communication.
- Teamwork.
- Creativity in proposing imaginative and original solutions.
- Organization.
- STEAM skills

Workshop 3: The women in the aerospace sector

Key Features

- Age range: 8–14 years
- Type: group activities
- Difficulty: moderate
- Duration: 1 hour 30 minutes
- Materials included: folder, construction paper, pencils, rubbers, and crayons

Summary

The main objective of this activity is to think about stereotypes in STEAM professions and make women in aerospace science and technology more visible. The goal is to inspire and motivate girls to pursue their dreams and help boys break down their limiting beliefs and share this message.

Two main activities will take place in this workshop. The first will focus on thinking about and identifying unconscious behaviors and biases regarding the role of women in the aerospace sector. The second activity will involve playing a card game to learn about the career paths and professional lives of some important women in the sector, thereby creating role models.

The students will learn...

- Recognizing one's own biases and stereotypes.
- Factors that lead to marginalization, discrimination, and social injustice in the local community and around the world.
- Valuing equal rights for men and women in all areas of life.
- Recognizing differences based on birth, race, gender, opinion, or any other personal or social condition or circumstance as an enriching element of interpersonal relationships.

Skills to be developed...

- Communication.
- Recognizing one's own biases and stereotypes.
- Awareness of equal social rights.

5 GENDER EQUALITY

10 REDUCED INEQUALITIES

Workshop 4: Constellations as a life notebook

Key Features

- Age range: 6–14 years
- Type: group activities
- Difficulty: moderate
- Duration: 1 hour 30 minutes
- Materials included: folder, pencils, erasers, and colored pencils

Summary

The main objective of this activity is to understand that the night sky has played—and continues to play—a key role in human life, whether in understanding the passage of time, organizing agricultural activities, or knowledge transfer.

Depending on the academic year this workshop is held, there will be either one or two main activities. The first establishes the basics and focuses on understanding the night sky as a medium for transmitting knowledge through a story specially prepared for the workshop, followed by the writing of a story of their own. The second activity involves identifying the constellations studied and creating their own. In this way, by combining both activities, we can adapt to the educational requirements of each grade level we are addressing.

The students will learn...

- Identifying and locating the most prominent constellations in the night sky of the Northern Hemisphere.
- Areas of human life in which the night sky has been used as a guide.
- How the night sky changes throughout the year.
- Traditional methods of knowledge transfer.
- Creativity in designing your own constellations and assigning meaningful stories to them.

Skills to be developed...

- Communication.
- Creativity in writing and storytelling.
- Teamwork.
- Orientation on a world map.

4 QUALITY EDUCATION

13 CLIMATE ACTION

Workshop 5: Biology in space: What do aliens look like?

Key Features

- Age range: 9–14 years
- Type: group activities
- Difficulty: moderate
- Duration: 1 hour 30 minutes
- Materials included: folder, pencils and drawing supplies, and planet cards.

Summary

The main objective of this activity is to explore astrobiology and understand what life on other planets might be like based on their environments. Through a theoretical introduction and a hands-on activity, participants will discover the key concepts of the search for extraterrestrial life and how environmental conditions determine the evolution of living beings.

The workshop combines interactive explanations of how we search for life beyond Earth and how aliens are represented in popular culture with a hands-on activity in which students will work in groups to imagine and design an alien creature based on the characteristics of a fictional planet. This structure allows the workshop to be adapted to different educational levels, promoting critical thinking, creativity, and teamwork. Materials included: workshop booklet, pencils, drawing supplies, and planet cards.

The students will learn...

- The basic principles of astrobiology and their relationship to the search for extraterrestrial life.
- The scientific methods used to detect life on other planets.
- The influence of environmental conditions on the evolution of living organisms.
- Differences between the portrayal of aliens in popular culture and actual scientific possibilities.
- Teamwork and creativity in designing an alien being adapted to a specific environment.
- Scientific reasoning and justification in the presentation of their proposals.

Skills to be developed...

- Communication.
- Creativity in designing and explaining alien beings.
- Teamwork.
- Critical thinking and scientific reasoning.

4 QUALITY EDUCATION

14 LIFE BELOW WATER

15 LIFE ON LAND

Workshop 6: Create your own star system!

Key Features

- Age range: 11–14 years
- Type: group activity
- Difficulty: moderate
- Duration: 2 hours
- Materials included: pencil, rubbers and colored pencils

Summary

Based on the school curriculum for the subject of environmental studies, this workshop reinforces key concepts and the more abstract aspects of the topic.

The main objective of this activity is to help students internalize concepts such as the origin of the universe and our place within it, our family within the solar system, the Earth's movements, the consequences of these movements, the dance with our natural satellite, the Moon, and its effects, the reasons behind Earth's climates and seasons, and raising awareness about the importance of caring for the environment.
By setting up a scenario where students are the first explorers of a new star system very similar to our solar system, the class is divided into teams that must correctly identify all the variables of the system and the planet most likely to support life.

The students will learn...

- The origin of the universe.
- Types of celestial bodies found in space.
- Our place in the universe, the Milky Way, and the Solar System.
- Earth's movements and their consequences.
- The Moon, our natural satellite, and its effects on us.
- Earth's climates and seasons.
- Environmental awareness.

Skills to be developed...

- Communication.
- Teamwork.
- Division of tasks to collaborate as part of a large team.
- Creativity and problem-solving skills to imagine new worlds.

Challenge AulaSputnik

Key Features

- Age range: 8–16 years
- Type: group activity
- Difficulty: moderate
- Duration: 6–7 sessions, each lasting 1.5 hours.
- Materials included: an educational kit per workshop, LEGO building blocks, school supplies, and 3D-printed parts

Summary

Are you looking for a three-month project that integrates various STEAM activities while encouraging curiosity, creativity, and teamwork?

The AulaSputnik Challenge is the perfect activity for you!

A unique educational experience in the format of a space mission, where your students will take the lead in their own collaborative project, working as a team over several weeks to achieve a common goal.
Through interconnected workshops following a common narrative, each session builds on knowledge that is directly applied to their final space project, achieving sophisticated and valuable results for the students... and for the school as well!
Ideal for teachers looking to transform their classroom into a laboratory of ideas, collaboration, and discovery.

The students will learn...

- Constellations as a Life Notebook
- Biology in Space: What Do Aliens Look Like?
- How Do Nanosatellites Help Our Planet?
- Explore Space with AulaSputnik Rockets!
- Create your own star system!
- Egg drop (for 12-16-year-olds only)
- Women in the aerospace sector (addressed across all workshops)

Skills to be developed...

- Communication.
- Teamwork and contributing to a larger project.
- Long-term project development.
- Creativity and problem-solving skills

Feedback from the workshops

Students' answers to the question:

What have you learned in the AulaSputnik workshops?

- “NANOSATELLITES CAN HELP THE WORLD.”
- “WHAT IS STEAM.”
- “MORE WOMEN SHOULD WORK IN STEAM.”
- “YOU CAN PURSUE WHAT YOU LOVE, WHETHER YOU’RE A MAN OR A WOMAN.”

Teachers' responses to the question:

What has been the most meaningful learning experience for you from the AulaSputnik workshops?

- How the STEAM concept is part of our daily lives without us even realizing it.
- The most meaningful lesson has been learning about women in the world of science and the value of women’s work in various fields.
- Reflecting on our behavior and becoming aware of our attitudes.

Space inspires us and makes us dream!

www.aulasputnik.com

aulasputnik@gmail.com

@aulasputnik

@aula_sputnik